# Mr Munch

AI DINING CONCIERGE

# Eat out without the fear

Your AI dining concierge for allergies and diets

Armando Medina · Marco Germanier · Encode 2026



# Eating out shouldn't be a gamble

**2.4M**

UK adults live with a clinically confirmed food allergy

Food Standards Agency, 2024

**1 in 100**

people in the UK have coeliac disease

Coeliac UK

**3×**

rise in food-anaphylaxis hospital admissions, 1998–2018

BMJ, 2021

*Menus rarely list allergens. Every dinner starts with phone calls, cryptic menus and guesswork. In a group, everyone has a different rule.*

# From craving to a confirmed table



### Find

Nearby places from Google Maps, cross-checked with OpenStreetMap diet tags

### Read

Menus from restaurant websites and photos of specials boards

### Check

Every dish against the UK's 14 major allergens and your diets

### Book

Your Saint agent books the table, after asking you first

You say *“Dinner for two near Shoreditch. One vegan, one shellfish allergy.”*

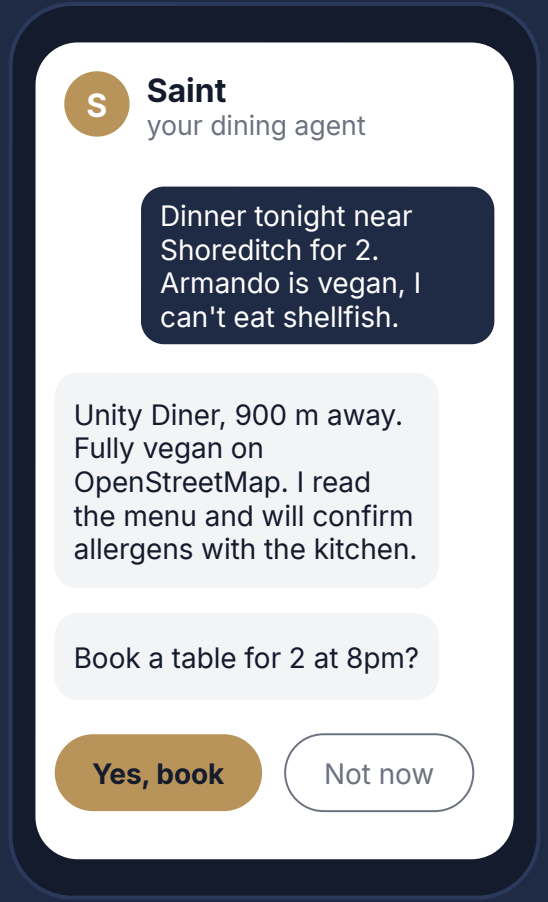
LIVE DEMO

# Let's book dinner

> *“Find us dinner tonight near Shoreditch. Armando is vegan, Marco can't eat shellfish.”*

## Watch for

- Atlas hybrid search recalls dishes it already knows
- Google Maps + OpenStreetMap find vegan-friendly places
- Menus read and checked against 14 allergens
- Saint asks before it books



# MongoDB Atlas is the shared brain

One platform for data, search, embeddings and reranking. No stitching.

**Menu read** › **Auto-embed voyage-4** › **Text + vector $rankFusion** › **Voyage rerank** › **Grounded answer**

### Atlas Search
Exact words: prawn, gluten

### Vector Search
Finds dishes by meaning

### Automated Embedding
Atlas embeds every dish itself

### Hybrid Search
$rankFusion, allergens out

### Voyage AI Rerank
rerank-2.5 puts the best first

### Stream Processing
Menu photos as events (next)

**630+**

dishes from real menus, shared by every user

*Every menu one person reads makes the next search smarter.*

# MongoDB features in our code

### Atlas Search

Exact allergen words, typos

```
$search: { compound: {
  must: [{ text: { query,
    fuzzy: { maxEdits: 1 } } }],
  mustNot: [{ equals: {
    path: "allergensPresent",
    value: "crustaceans" } }] } }
```

### Vector Search + Auto-embed

Meaning; Atlas embeds every dish

```
{ type: "autoEmbed",
  path: "searchText",
  model: "voyage-4" }
$vectorSearch: {
  query: { text: "vegan" },
  filter: { allergensPresent:
    { $nin: avoid } } }
```

### Hybrid Search

One ranked list from both

```
$rankFusion: {
  input: { pipelines: {
    vectorPipeline,
    fullTextPipeline } },
  combination: { weights: {
    vectorPipeline: 0.55,
    fullTextPipeline: 0.45 } } }
```

### Voyage AI rerank

Best dishes first, then the LLM

```
{ $rerank: {
    model: "rerank-2.5",
    path: "searchText",
    numDocsToRerank: 40 } }

// Voyage API on MongoDB 8.0
```

### Geospatial

Restaurants and dishes near you

```
{ location: "2dsphere" }

location: { $near: {
  $geometry: { type: "Point",
    coordinates: [lng, lat] },
  $maxDistance: 2000 } }
```

### Upserts + unique index

Shared knowledge, no duplicates

```
createIndex(
  { restaurantPlaceId: 1,
    name: 1 },
  { unique: true })

updateOne(key, { $set: dish },
  { upsert: true })
```



# Honest, private and in your control

## Honest

Unknown never counts as safe. Each place gets a confidence level, or a nudge to ask the kitchen.

## Private

We never store anyone's allergies. Your own Saint keeps them and shares them per request.

## In your control

Saint asks before it sends a message, books a table or pays a deposit.



# Eat out without the fear

**Armando Medina** Backend · @armsves

**Marco Germanier** Data · @0xr3x

mrmunch.vercel.app · github.com/0xr3x/mr-munch

Photos: StockSnap (CC0)